

Personalized AI/ML Learning Roadmap

Contents

1	Introduction	2
2	Phase 1: Foundations in Programming & Math (Months 1–3)	2
2.1	Objectives	2
2.2	Core Curriculum	2
2.3	Hands-On Projects	2
2.4	Success Metrics	2
3	Phase 2: Introduction to Machine Learning (Months 4–6)	2
3.1	Objectives	3
3.2	Core Curriculum	3
3.3	Hands-On Projects	3
3.4	Success Metrics	3
4	Phase 3: Intermediate ML and Deep Learning (Months 7–9)	3
4.1	Objectives	3
4.2	Core Curriculum	3
4.3	Hands-On Projects	4
4.4	Success Metrics	4
5	Phase 4: Advanced Topics and Specialization (Months 10–12)	4
5.1	Objectives	4
5.2	Core Curriculum	4
5.3	Hands-On Projects	4
5.4	Success Metrics	5
6	Specialization Track: Generative AI & LLMs	5
7	Portfolio Building Guidance	5
8	Career Preparation	5
9	Continuous Learning Strategy	5

1 Introduction

This roadmap is designed for a complete beginner with no coding background, aiming to become an ML Engineer specializing in Generative AI & LLMs. With a commitment of 15–20 hours/week, the timeline spans 12–18 months. The approach emphasizes project-driven learning. Adjust the context by providing specific details for a tailored revision.

2 Phase 1: Foundations in Programming & Math (Months 1–3)

Duration: 3 months (180–240 hours total).

2.1 Objectives

Master Python programming and essential math (linear algebra, calculus, probability, statistics) for ML. Build problem-solving and data-handling skills.

2.2 Core Curriculum

- Online Courses:
 - “Python for Everybody” on Coursera (University of Michigan) – Free, beginner-friendly.
 - “Mathematics for Machine Learning” Specialization on Coursera (Imperial College London) – Covers vectors, matrices, derivatives, and stats.
- Textbooks:
 - Python Crash Course by Eric Matthes – Hands-on Python basics.
 - Mathematics for Machine Learning by Marc Peter Deisenroth et al. – Free PDF, ML-focused math.
- Tools/Libraries: Python, Jupyter Notebooks, NumPy (arrays, basic math).

2.3 Hands-On Projects

1. Dataset Analysis: Python script to compute averages and plot charts using NumPy and Matplotlib.
2. Math Simulator: Functions for matrix multiplications or probability distributions (e.g., coin flips).
3. Budget Tracker: Console app for spending calculations and visualization.

2.4 Success Metrics

- Write and debug a 100-line Python program independently.
- Explain derivatives and their role in optimization.
- Achieve 80%+ accuracy on course quizzes and run projects error-free.

3 Phase 2: Introduction to Machine Learning (Months 4–6)

Duration: 3 months (180–240 hours total).

3.1 Objectives

Understand supervised/unsupervised learning, regression, classification, and evaluation metrics. Gain experience in data preprocessing and simple models.

3.2 Core Curriculum

- Online Courses:
 - “Machine Learning” by Andrew Ng on Coursera – Covers fundamentals with Python labs.
 - “Practical Deep Learning for Coders” by Fast.ai – Free, project-focused.
- Textbooks:
 - Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurélien Géron
 - Practical code examples.
- Tools/Libraries: Scikit-Learn, Pandas, Matplotlib/Seaborn.

3.3 Hands-On Projects

1. Iris Classification: Scikit-Learn model to predict flower species; evaluate accuracy.
2. House Price Prediction: Regression on Kaggle dataset (e.g., Boston Housing).
3. Sentiment Analysis: Text classification using bag-of-words and logistic regression.

3.4 Success Metrics

- Explain overfitting, cross-validation, and gradient descent.
- Build a model achieving 80%+ test accuracy.
- Submit a Kaggle competition entry and understand rankings.

4 Phase 3: Intermediate ML and Deep Learning (Months 7–9)

Duration: 3 months (180–240 hours total).

4.1 Objectives

Learn neural networks, CNNs, RNNs, and deep learning frameworks. Master model optimization and hyperparameter tuning.

4.2 Core Curriculum

- Online Courses:
 - “Deep Learning Specialization” by Andrew Ng on Coursera – Neural nets, CNNs, sequence models.
 - “Introduction to TensorFlow for AI, ML, and Deep Learning” on Coursera – TensorFlow hands-on.
- Textbooks:

- Deep Learning by Ian Goodfellow et al. – Reference for deeper understanding.
- Tools/Libraries: TensorFlow or PyTorch (PyTorch preferred for Generative AI), Keras.
- Key Papers: “A Few Useful Things to Know About Machine Learning” by Pedro Domingos.

4.3 Hands-On Projects

1. Image Classifier: CNN in PyTorch for MNIST digit classification.
2. Stock Price Predictor: RNN/LSTM for time-series forecasting.
3. Chatbot Prototype: Simple sequence model for text generation.

4.4 Success Metrics

- Build and explain a CNN or RNN, including backpropagation.
- Improve model performance by 10–20% via tuning.
- Deploy a model to a web app (e.g., Streamlit).

5 Phase 4: Advanced Topics and Specialization (Months 10–12)

Duration: 3 months (180–240 hours total).

5.1 Objectives

Master transfer learning, deployment, and ethics. Specialize in Generative AI & LLMs, focusing on transformers and fine-tuning.

5.2 Core Curriculum

- Online Courses:
 - “Generative AI with Large Language Models” by DeepLearning.AI on Coursera.
 - “MLOps Specialization” on Coursera – Production deployment.
- Textbooks:
 - Designing Machine Learning Systems by Chip Huyen – Scalable ML.
- Tools/Libraries: Hugging Face Transformers, Docker, MLflow.
- Key Papers: “Attention Is All You Need” (Transformer); “GPT-3: Language Models are Few-Shot Learners”.

5.3 Hands-On Projects

1. Fine-Tune LLM: Adapt a pre-trained model (e.g., BERT) for text generation.
2. Generative Image Model: Build a GAN or use Stable Diffusion.
3. ML Pipeline: Deploy a text summarizer on AWS/Heroku.

5.4 Success Metrics

- Fine-tune an LLM, achieving task-specific benchmarks (e.g., BLEU score).
- Deploy a model and handle edge cases.
- Summarize 2–3 papers, applying concepts to a project.

6 Specialization Track: Generative AI & LLMs

Focus on transformer-based models. Dedicate 50% of Phase 4 to Hugging Face and prompt engineering. Explore multimodal models (e.g., CLIP). Alternatives: Pivot to Computer Vision (OpenCV projects) or Reinforcement Learning (Gym library). For ML Engineering, emphasize MLOps (e.g., Kubeflow).

7 Portfolio Building Guidance

- GitHub Repos: Create 10+ repos with READMEs detailing problem, approach, code, results. Use Jupyter notebooks.
- Kaggle: Join 5+ competitions (e.g., Titanic). Aim for top 20% in intermediate ones.
- Blogs/Site: Write 4–6 Medium posts on projects. Host portfolio on GitHub Pages.
- Other: Contribute to open-source ML libraries. Record demo videos on YouTube.

8 Career Preparation

- Job Applications: Tailor resumes to highlight projects (e.g., “Deployed LLM using PyTorch, improved generation quality by 15%”). Apply to entry-level ML Engineer roles via LinkedIn/Indeed (10/week).
- Interviews: Practice ML theory (bias-variance), coding (LeetCode), and system design (e.g., recommendation system). Use Pramp for mocks. Item’s Networking: Join Reddit (r/MachineLearning), LinkedIn groups, NeurIPS meetups. Connect with 5 professionals/week.

9 Continuous Learning Strategy

Dedicate 2–4 hours/week post-roadmap:

- Research/Trends: Follow Stanford AI Index and ArXiv via Towards AI. Subscribe to The Batch newsletter.
- Frameworks/Tools: Experiment with PyTorch updates, Hugging Face betas.
- Community: Join hackathons (Devpost), podcasts (Lex Fridman), and learn tools like LangChain.
- Ethics: Study AI Ethics Guidelines (e.g., Google).